

AI-Powered Credit Risk Intelligence Platform

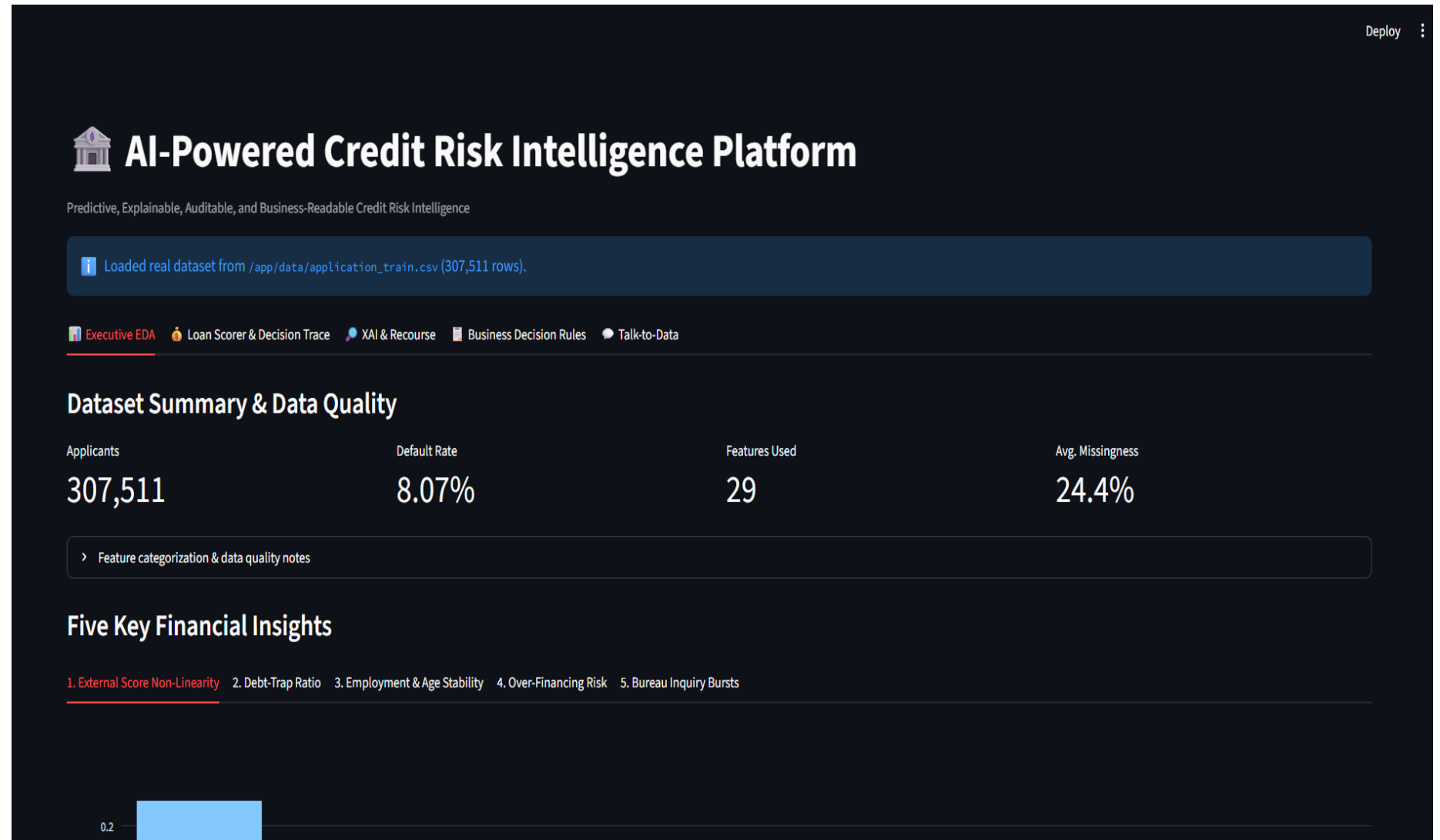
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Problem Statement

- Credit institutions need to evaluate whether a loan applicant is likely to be a low-risk or high-risk borrower.
- **Proposed Solution**
- An **AI-powered Credit Risk Intelligence Platform** that analyzes applicant and financial information to:
 - Predict credit risk
 - Identify important risk factors
 - Provide explainable decisions
 - Support real-time loan assessment
 - Assist business decision-making

Dashboard & Dataset Overview

The dashboard is with an overview of the dataset and its quality. The dataset contains **307,511 applicants, 29 features, an 8.07% default rate, and approximately 24.4% average missingness**, providing the foundation for the subsequent analysis.



Financial insight – External score non linearity

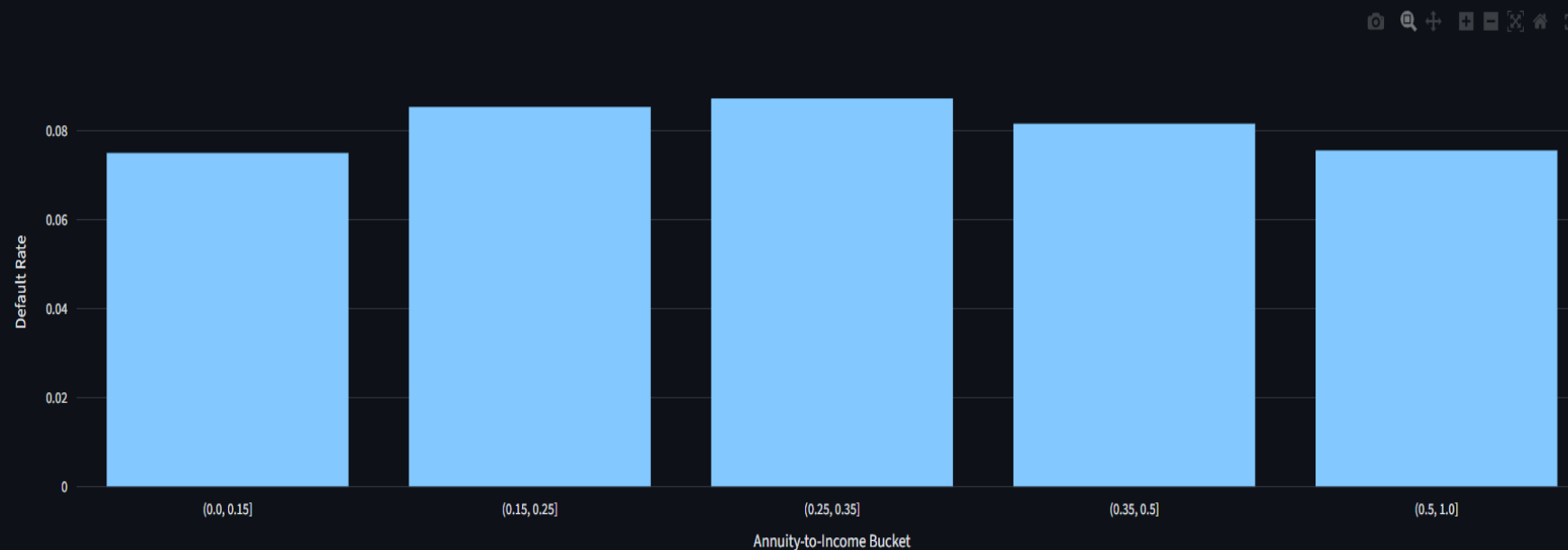
The analysis shows that default risk changes sharply as the external bureau score increases rather than following a simple linear relationship. Applicants with lower external scores have substantially higher default rates, making this one of the strongest indicators of credit risk.



Financial insight – Debt-to-Income

Five Key Financial Insights

1. External Score Non-Linearity 2. Debt-Trap Ratio 3. Employment & Age Stability 4. Over-Financing Risk 5. Bureau Inquiry Bursts



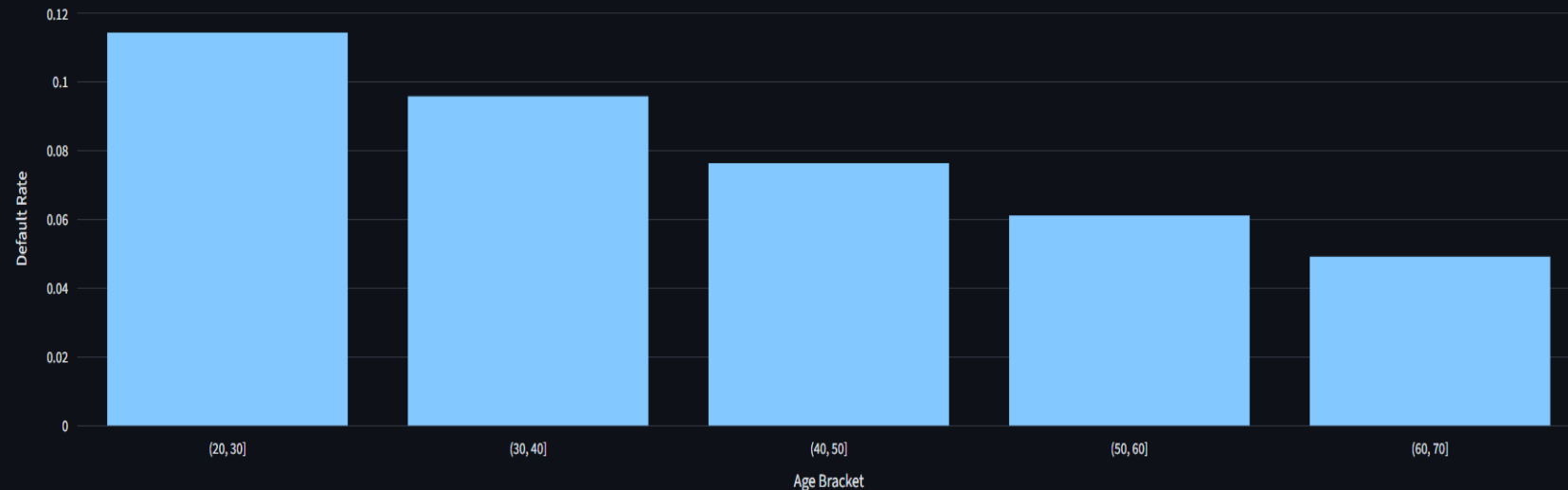
Default risk climbs once the annuity-to-income ratio crosses ~35% — the threshold used in the hard knockout rule (Tab 4).

The relationship between the annuity-to-income ratio and loan default risk. Default risk increases noticeably when the ratio crosses approximately **35%**, highlighting the importance of affordability and excessive repayment burden.

Financial insight – Employment and age Stability

Five Key Financial Insights

1. External Score Non-Linearity 2. Debt-Trap Ratio 3. Employment & Age Stability 4. Over-Financing Risk 5. Bureau Inquiry Bursts



Younger applicants (20-30) show a materially higher default rate than older, more employment-stable brackets.

Age and employment stability are analyzed as indicators of borrower reliability. The results show that younger applicants, particularly those in the **20–30 age range**, have a materially higher default rate than older and more employment-stable applicants.

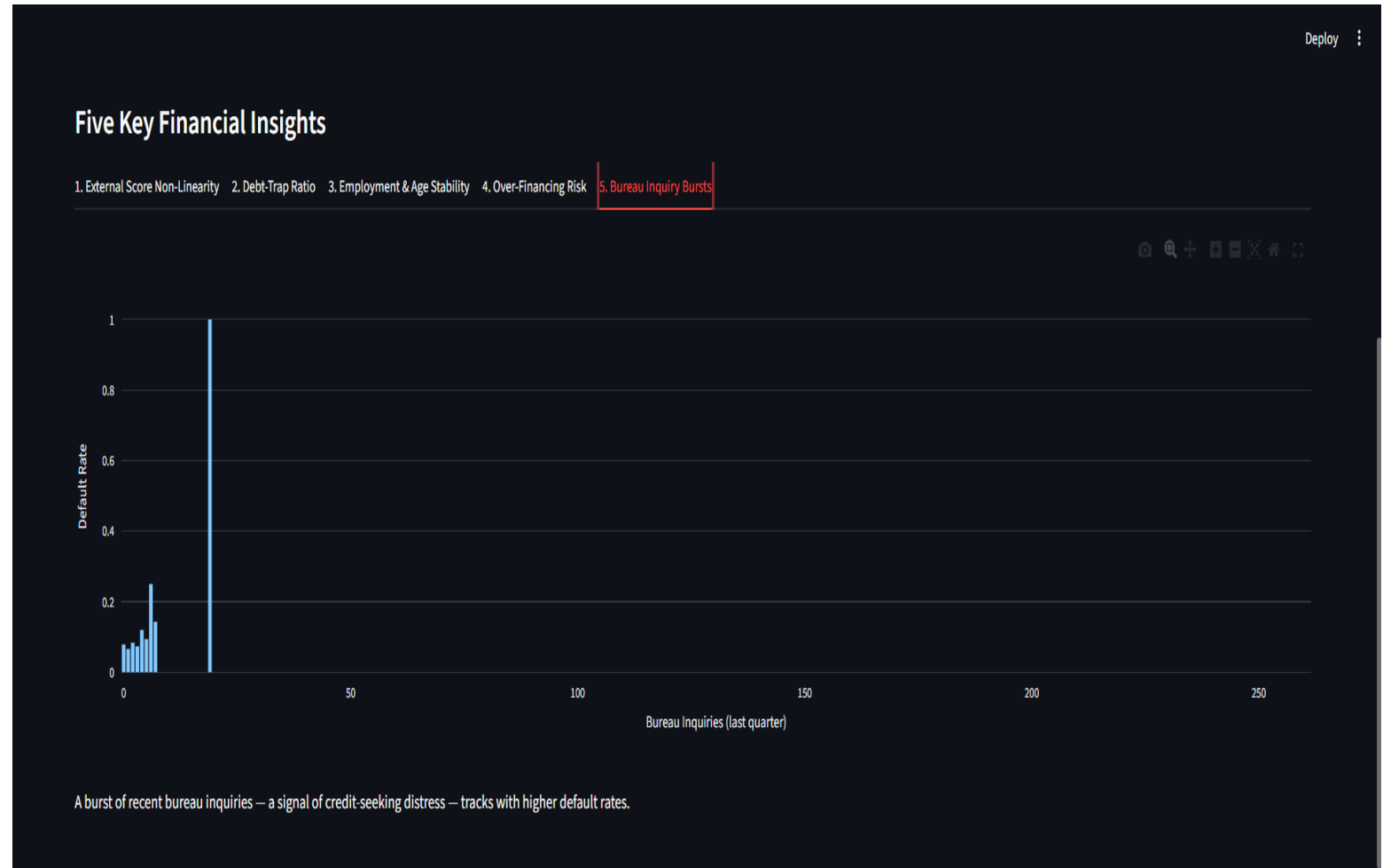
Financial insight – Over Financing risk

This analysis focuses on the relationship between the requested credit amount and the price of the purchased goods. Applicants requesting credit above the goods price demonstrate higher default risk, indicating potential over-financing or under-collateralized lending.



Financial insight – Bureau Inquiry Bursts

Recent bureau inquiries are used as an indicator of increased credit-seeking behavior. A higher number of inquiries within the last quarter is associated with greater default risk and can therefore act as an early warning signal.



Real Time Loan Scorer

Real-Time Loan Scorer

Load a sample applicant, or fill the form manually

Prime applicant

Annual Income

319500.00

-

+

Age (years)

38

Requested Credit Amount

1431000.00

-

+

Years Employed

5

Annuity (periodic payment)

39483.00

-

+

External Bureau Score (mean, 0-1)

0.65

Goods Price

1431000.00

-

+

Family Members

2

Income Type

Working

Education

Secondary / secondary special

Contract Type

Cash loans

Bureau Inquiries (last quarter)

1

Score Applicant

Probability of Default

12.8%

AI Credit Risk Score ?

780

Risk Band

Low

Expected Loss ?

\$82,141

No Tier 1 hard-knockout rules triggered.

The dashboard converts the applicant's information into an actionable risk assessment. For the demonstrated applicant, the system reports a **12.8% probability of default**, a **Low risk band**, an **AI Credit Risk Score of 780**, and an **expected loss of \$82,141**, while also checking policy rules.

Global Feature Importance



AI-Powered Credit Risk Intelligence Platform

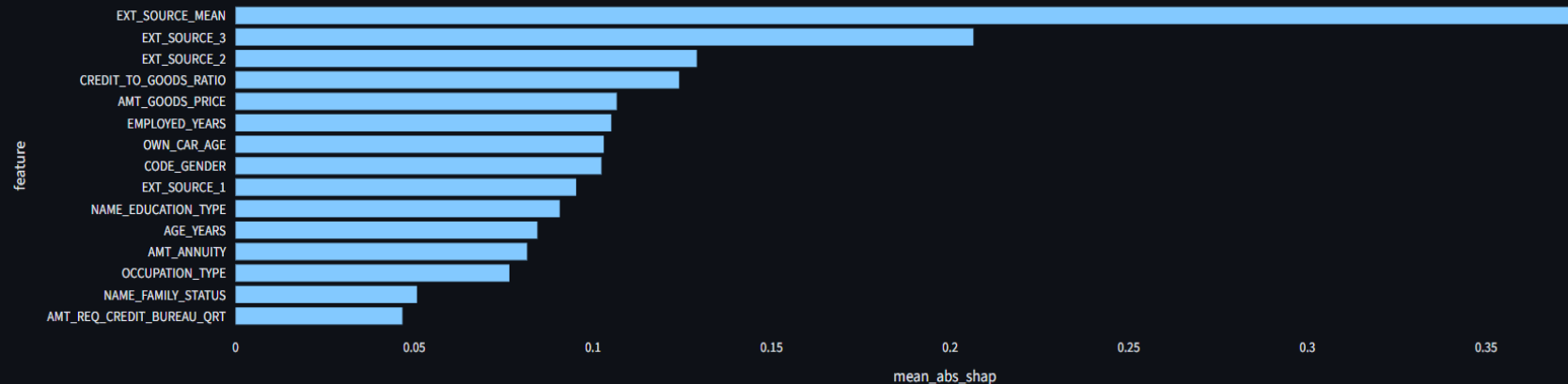
Predictive, Explainable, Auditable, and Business-Readable Credit Risk Intelligence

Loaded real dataset from /app/data/application_train.csv (307,511 rows).

Executive EDA Loan Scorer & Decision Trace **XAI & Recourse** Business Decision Rules Talk-to-Data

Global Feature Importance

Top 15 Global Risk Drivers (mean |SHAP|)

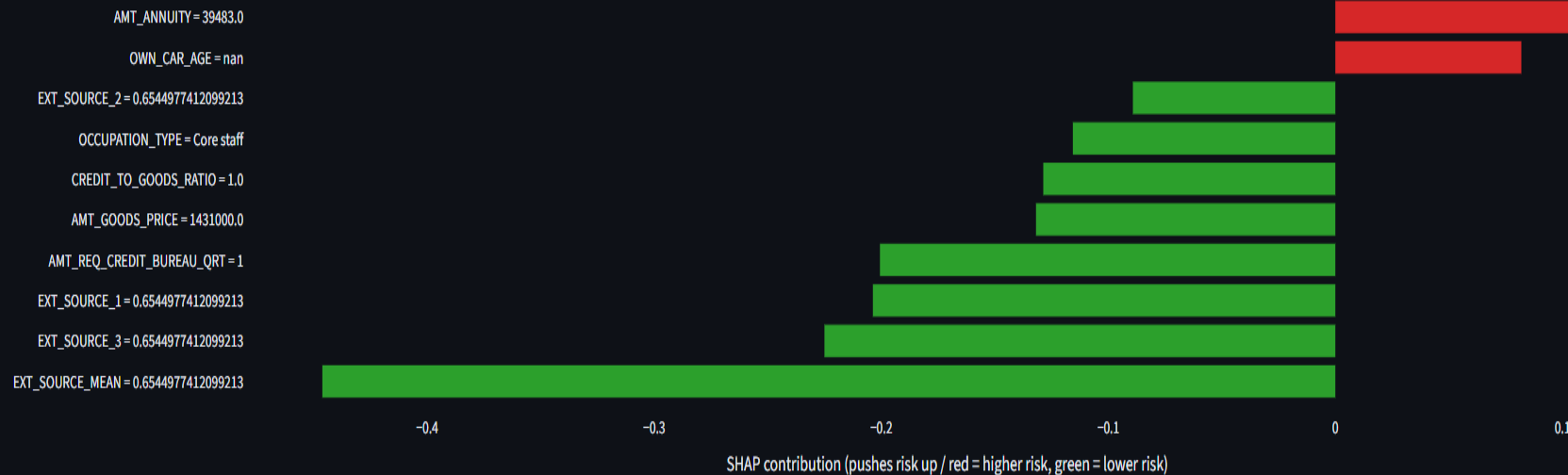


The dashboard shows which variables have the greatest overall influence on the model's predictions. Global SHAP feature importance provides transparency by showing the major risk drivers across the applicant population rather than focusing on only one individual.

Local Explanation & Counterfactual Recourse

Local Explanation & Counterfactual Recourse

Top drivers for this applicant (base log-odds: -0.48)



Recourse simulator applies to High Risk applicants.

The platform provides applicant-level explanations using SHAP contributions to show why a particular prediction was made. It also includes counterfactual recourse, allowing users to understand what changes to an applicant's characteristics could potentially improve their risk outcome.

Tier 1: Hard Knockout Business Rules

[Deploy](#)



AI-Powered Credit Risk Intelligence Platform

Predictive, Explainable, Auditable, and Business-Readable Credit Risk Intelligence

 Loaded real dataset from `/app/data/application_train.csv` (307,511 rows).

 Executive EDA  Loan Scorer & Decision Trace  XAI & Recourse  **Business Decision Rules**  Talk-to-Data

Business Decision Rules

Human-readable credit policy logic, bridging the trained ML model and raw dataset statistics for review by non-technical credit risk officers.

Tier 1 — Hard Knockout Rules (Policy Gate)

Fixed business thresholds. Non-negotiable — applied before the model is even consulted.

- IF Annuity-to-Income ratio > 50% THEN REJECT (unaffordable debt-service obligation)
- IF Income Type = 'Unemployed' THEN REJECT (no verified income source)
- IF Credit Amount > 150% of Goods Price THEN REJECT (severe over-financing / under-collateralized lending)
- IF mean External Bureau Score < 0.1 THEN REJECT (critically low third-party credit signal)

7,684 of 307,511 applicants in the current dataset would trigger at least one Tier 1 knockout.

Tier 2: Model-Derived Business Rules

Tier 2 — Model-Derived Business Rules (surrogate decision tree)

A shallow decision tree fit on the trained model's own High-Risk calls, read out as IF-THEN rules and audited against real outcomes.

View as:

- ☐ Plain-English bullets
- ☒ Structured table
- ☐ Raw decision tree (sklearn.tree.export_text)



IF (condition)	Support (n)	Coverage (%)	Precision (%)
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN <= 0.25 AND EXT_SOURCE_3 <= 0.50 AND AGE_YEARS <= 59.66	1757	2.86	88.79
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN > 0.25 AND CREDIT_TO_GOODS_RATIO > 1.21 AND AGE_YEARS <= 50.37	846	1.38	68.09
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN <= 0.25 AND EXT_SOURCE_3 > 0.50 AND AGE_YEARS <= 50.37	1054	1.71	63.47
EXT_SOURCE_MEAN <= 0.32 AND EXT_SOURCE_MEAN > 0.25 AND CREDIT_TO_GOODS_RATIO <= 1.21 AND EXT_SOURCE_3 <= 0.50	647	1.05	50.85

Talk-to-Data Assistant

Deploy

Executive EDALoan Scorer & Decision TraceXAI & RecourseBusiness Decision RulesTalk-to-Data

Talk-to-Data Assistant

Any API key (optional — leave blank to use offline verified query library)

Force offline query-library mode

Current mode: Offline Query Library

> Prompt engineering: semantic schema dictionary & token reduction

Try a question, or pick one of the 5 verified queries below:

Verified query library

-- free text below instead --

Or ask your own question in plain English

Ask

AI Credit Risk Score is a presentation-layer transform of the model's predicted default probability and is not a FICO score or regulated credit score.

The Talk-to-Data Assistant provides a conversational interface for interacting with the credit-risk dataset. Users can select verified queries or ask questions in plain English, reducing the need to manually construct database queries or perform technical analysis.

Conclusion

- **Predictive:** Identifies applicants with higher probability of default.
- **Explainable:** Reveals the key factors behind every risk decision.
- **Actionable:** Combines AI predictions with business rules and real-time scoring.
- **Intelligent:** Enables natural-language interaction with credit data.
- **The platform transforms credit data into transparent, actionable intelligence for faster and smarter lending decisions.**

THANK YOU !!!!